Dongheon Lee

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Profile

AI researcher specializing in **computer vision**, **vision-language models**, and **data-efficient learning**. Experienced in few-shot and incremental learning, multimodal learning, vision-language model adaptation, and attention mechanisms for vision transformer.

Education

M.S. in Artificial Intelligence Mar. 2024 – Aug. 2026
Chung-Ang University *GPA: 4.13/4.5*
Advisor: Joonki Paik
Thesis Title: Logit-Calibrated Differential Attention for Vision Transformers

B.S. in Astronomy & Space Science Mar. 2016 – Aug. 2023
Kyung Hee University *GPA: 3.54/4.5*

Publications

C: Conference / Journal, **W**: Workshop, **U**: Under Review *: Equal Contribution, †: Corresponding Author

Journal & International Conference

[C1] PLuG: Pairwise Logit Gating for Expressive Attention Modulation in Vision Transformers Paper
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik†
IEEE Access, 2026

[C2] NAPP: Noise-Adaptive Prototype Perturbation for Few-Shot Learning Paper
Il Hwan Kim, Sang Woo Yun, **Dongheon Lee**, Kim Seong Su, and Joonki Paik†
IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026

[C3] CEAL: Cross-Expert Attention with Curated Peer Selection for Scalable and Privacy-Preserving Exemplar-Free Continual Learning Paper
Seongsu Kim*, **Dongheon Lee***, Sangwoo Yun, Ilhwan Kim, and Joonki Paik†
IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026

[C4] ViSeNet: A Visual-Semantic Fusion Network for Enhancing Few-Shot Classification Paper
Dongheon Lee, Sangwoo Yun, and Joonki Paik†
IEEE International Conference on Consumer Electronics (ICCE), 2025

Workshop Papers

[W1] Unified Visions: Multi-Modal Transformer Fusion for Comprehensive VQA Answering
Minju Baek*, **Dongheon Lee***, Jaehong Yoon, and Joonki Paik†
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) VizWiz Workshop, 2025

[W2] Test-Time Augmented Ensemble for Zero-Shot Learning
Dongheon Lee*, Minju Baek*, Jaehong Yoon, and Joonki Paik†
IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) VizWiz Workshop, 2025

Research in Progress

[U1] RoME: Routed Multi-Expert LoRA for Few-Shot CLIP Adaptation
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik†

[U2] Structured Adaptation and Subspace Regularization for Few-Shot Vision-Language Learning
Dongheon Lee, Sangwoo Yoon, Seongsu Kim, and Joonki Paik†

[U3] STAR: Semantic-to-Temporal Anchored Refinement for Open-Vocabulary Audio-Visual Event Localization
Dongheon Lee, Jihun Park, Yerin Lim, and Joonki Paik†

[U4] Logit-Calibrated Differential Attention for Vision Transformers
Dongheon Lee and Joonki Paik†

Research & Project Experience

Dronebot Combat System Response Next-Generation Intelligent Surveillance and Reconnaissance Mission Equipment 2024 – 2026

Funded by Korea Research Institute for Defense Technology Planning and Advancement (KRIT)

- Developed cross-domain few-shot object detection methods for surveillance scenarios with fewer than 10 training images per class and a severe base-to-novel domain gap.
- Explored multimodal object recognition by incorporating LLM-based textual class semantics and higher resolution scaling to improve recognition of novel objects, improving detection performance from approximately 40 AP to 75 AP.

Location-Based AI Image Interpretation Platform for Sewer Pipe Defects 2024 – 2026

Funded by Ministry of SMEs and Startups

- Served as the project lead and developed YOLO-based pipelines for sewer pipe defect instance segmentation and video-based analysis.
- Improved dataset quality through relabeling, augmentation, and preprocessing, expanding the dataset from 5,300 to 30,000 images and improving performance from 0.45 to 0.70 mAP.

Multimodal Generative AI for Multidimensional Scene Analysis and Reproduction 2025

Funded by National Research Foundation of Korea (NRF)

- Investigated All-In-One Image Restoration models.

Technical Skills & Languages

Programming	Python
Frameworks	PyTorch, timm
AI / ML	Computer Vision, Few-Shot Learning, Incremental Learning, Multimodal Learning, Vision-Language Model Adaptation, and Attention Mechanism
Language	English (OPIc IH)

Patents

Session Class Prototype Incremental Learning (SCPIL): Mitigating Catastrophic Forgetting with Distance-Based Prototype Learning. 10-2935903